PADHAKU
Curiosity · Class 6 Science



CHAPTER 8

A Journey through States of Water

NCERT Exercise Solutions

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✨ Try each question yourself first, then check the answer. 💡
answers are samples: write your own.

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⭐ Chapter opening

? **Aavi says ice and water are the same substance; Thirav says they are different because ice is hard and can be held in the hand. What do you think? Why?**

✔ ANSWER

Aavi is right. Ice and water are the **same substance** (water) in two different forms.

Ice is the **solid** form and water is the **liquid** form. Ice melts to give water, and water freezes to give ice.

? **Thirav thinks something gets added to water in the freezer to make ice. Do you think Thirav is right? How can you find out?**

✔ ANSWER

No, nothing gets added. Only cooling changes water into ice.

To find out: keep some water in a covered (sealed) container in the freezer, and take it out once it turns into ice. Let the ice melt at room temperature. We get back the same water, and its amount does not increase.

Activity 8.1 **Put an ice cube in a cup, leave it on the table and observe. What can you conclude? Does this mean that ice and water are the same substance?**

✔ ANSWER

Observation: The ice cube slowly melts and changes into water.

Conclusion: Yes, ice and water are the same substance. They are two different forms (**states**) of water.

⭐ 8.1 Investigating Water's Disappearing Act

? **Have you ever noticed water in the puddles disappearing? Where does it go? Where else have you seen water disappearing? Can you think of a possible reason why this happens?**

💡 SAMPLE ANSWER

Yes. Some water seeps into the soil and some goes into the air as water vapour.

Other examples: wet clothes drying, a wet floor drying after mopping, water drying from utensils, sweat drying on our body.

Possible reason: water changes into water vapour (**evaporation**), and some soaks into the ground.

? **Water left on the surface of washed utensils dries up after some time. Does the reason you thought earlier apply in this case also? Design an activity to investigate whether water seeps through the surface of the utensils.**

💡 SAMPLE ANSWER

Seeping into the soil cannot happen here, because steel does not let water pass. Only evaporation explains it.

Activity: put a spoon of water on a steel plate and check its lower side again and again until the water disappears (Activity 8.2). The lower side stays dry, so water does not seep through.

Activity 8.2 **Take a tablespoon of water on a steel plate. Observe whether water seeps through to the other side, at regular intervals, until the water disappears. What do you infer? Is this activity enough to conclude that water does not seep through a steel plate? Then where has the water gone?**

💡 SAMPLE ANSWER

Observation: The lower side of the plate stays dry. The water slowly disappears from the top.

Inference: Water does not seep through the steel plate.

To be more sure, we can repeat it many times, or keep a dry tissue paper under the plate and see that it stays dry.

The water has changed into its gaseous state, **water vapour**, and gone into the air.

Let us draw

While making dosa, we sprinkle some water on the hot pan and it disappears. Where does it go? Draw a detailed sketch (with labels and caption) about what happens to the water.

💡 SAMPLE ANSWER

The water changes into **steam** (water vapour) because of the heat of the pan and goes into the air. Some of it cools into tiny droplets, so we see white steam.

Sketch: a hot pan (label: hot pan) on a stove (label: flame), water drops on it (label: water sprinkled), and rising wavy lines (label: steam/water vapour going up). Caption: "Water on a hot pan changes into water vapour."

?

Can you think of other examples of evaporation?

✔ ANSWER

Drying of wet clothes, drying of a mopped floor, drying of sweat on our body, drying of wet hair, water drying from a puddle, drying of ink or paint.

? **What is the reason for the disappearance of water from the puddles? Is it due to: (i) seeping of water into the ground or (ii) evaporation of water or (iii) both of these?**

✔ ANSWER

(iii) Both of these. Some water seeps into the ground (soil lets water pass) and some evaporates into the air.

? **Hand sanitiser disappears as you rub it on your hands. What happens to it?**

✔ ANSWER

It **evaporates** quickly and goes into the air as vapour. That is why our hands also feel cool.

⭐ 8.2 Another Mystery

Activity 8.3 **Take cold water in a glass tumbler, add a few ice cubes and leave it for five minutes. Record your observations and questions in Table 8.1 (I observe / I wonder).**

💡 SAMPLE ANSWER

I observe	I wonder
Tiny water drops appear on the outer surface of the tumbler	Where do these drops come from?
Small drops join to form bigger drops	Is water seeping out of the glass?
The outer surface feels cold and wet	Will this happen with a metal tumbler also?

? **Suggest possible reasons explaining the appearance of water droplets on the outer surface of the glass tumbler. (Fig. 8.3)**

💡 **SAMPLE ANSWER**

- Water may have seeped out through the glass.
- Ice may have come out of the glass and melted.
- Water vapour in the air may have turned into water on the cold glass.

Fig. 8.4 **What do you think about the possible reasons (chain of reasoning) in Fig. 8.4? Continue the discussion or conduct activities to find evidence. Where else have you seen water droplets like this?**

💡 **SAMPLE ANSWER**

The idea of seeping is not correct: the water level does not go down, and a tumbler with water at room temperature stays dry outside. So the drops must come from the air.

Similar drops: dew drops on grass and leaves in the morning, drops outside a cold bottle taken from the fridge, mist on window glass in winter, drops under the lid of a cooking pot.

? **Why do we see dew drops more in the morning?**

✔ ANSWER

At night and early morning the air and the surfaces of leaves and grass become **cold**. Water vapour in the air touches these cold surfaces and **condenses** into dew drops.

? **When we boil water in a half-filled utensil and cover it with a steel plate, water drops collect on the inner side of the plate. Where do these water drops come from?**

✔ ANSWER

The boiling water changes into water vapour. The vapour touches the cooler steel plate and **condenses** into water drops.

? **Could the water appearing on the outer surface of the glass tumbler in Activity 8.3**

also be due to condensation of water vapour present in the air?

✔ ANSWER

Yes. The ice makes the glass surface cold. Water vapour in the air touches it and condenses into water droplets.

Activity 8.4

Take a glass tumbler half-filled with water and ice cubes, cover it with a steel plate and weigh it on a digital balance every five minutes for 30 minutes (Table 8.2). Predict: will the mass increase, decrease or remain the same? Do your findings match your predictions? Explain your observations.

💡 SAMPLE ANSWER

Prediction: The mass will increase.

Time	Mass (sample readings)
0 min	250.0 g
10 min	250.4 g
20 min	250.8 g
30 min	251.2 g

Observation: The reading slowly increases and water drops collect on the outside of the tumbler.

Explanation: Water vapour from the air condenses on the cold tumbler, so extra water is added and the mass goes up.

More to do

The amount of water vapour in the air is called humidity. Compile the daily humidity data of your area for the year and study any patterns.

💡 SAMPLE ANSWER

Humidity is usually **highest in the rainy season** (July–September) and lower in dry months like winter and summer (it also depends on the place, e.g. coastal areas stay humid).

Note the humidity from a newspaper or weather app every day and make a month-wise table.

?

Can we conclude that water is not seeping through the wall of the tumbler and that the water collected outside is only due to condensation? How would you modify Activity 8.4 to find the answer? What can you conclude?

✔ ANSWER

No, Activity 8.4 alone does not prove it.

Modification: mark the water level on the tumbler with a marker or tape and repeat the activity.

Observation: The water level does not go down, but water still collects outside.

Conclusion: Water is not seeping out; the extra water comes from **condensation** of water vapour in the air.

⭐ 8.3 What are the different States of Water?

Activity 8.5 **Move an ice cube between containers of different shapes, pour water between containers, pour water on a clean surface, and think how water vapour spreads. Complete Table 8.3. What are the differences in the properties of water in solid, liquid and gaseous states?**

✔ ANSWER

Property	Ice (Solid)	Water (Liquid)	Water vapour (Gas)
Shape	Fixed shape	No fixed shape; takes the shape of the container	No fixed shape
Ability to flow	Does not flow	Flows	Flows / moves freely
Ability to spread	Does not spread	Spreads (volume stays the same)	Spreads in the entire space available

? **Look around and find some examples of solid substances.**

✔ ANSWER

Stone, wood, glass, iron, brick, plastic, sugar, salt, ice.

? **Here are two examples of liquids— milk and oil. Think of five more examples.**

✔ ANSWER

Water, juice, kerosene, petrol, honey (also vinegar, coconut water).

? **You can smell food being cooked even without entering the kitchen. How does this smell reach us?**

✔ ANSWER

The smell is carried by gases that **spread through the air** in all the available space, and they reach our nose.

? **What are the other examples of gases you can think of?**

✔ ANSWER

Oxygen, carbon dioxide, nitrogen, water vapour, LPG (cooking gas), hydrogen, helium (in balloons).

⭐ 8.4 How can we change the States of Water?

? **How can you quickly change ice to its liquid state, water?**

✔ ANSWER

By **heating** it, e.g. keep it in the sun or warm it in a pan.

? **If we want to change water into ice, what should be done?**

✔ ANSWER

Place the water in a **cold place**, such as a freezer. The water freezes into ice.

? **How can we turn candle wax into liquid state? How can we change the liquid wax back into solid state?**

✔ ANSWER

Heat the wax (e.g. a lit candle) and it melts into liquid.

Cool the liquid wax and it becomes solid again.

? **What are the other liquids you have seen which get converted into a solid? Have you ever seen coconut oil getting converted into its solid state during winter?**

💡 SAMPLE ANSWER

Coconut oil and ghee become solid in winter; melted butter, melted chocolate and melted wax become solid on cooling.

Yes, coconut oil freezes (becomes white and solid) on cold winter days.

Activity 8.6 **Fill up the blank boxes A, B, C and 1, 2, 3, 4 in Fig. 8.5 using the words: Liquid, Freezes, Evaporates, Gas, Condenses. (Solid and Melts are filled.)**

✔ ANSWER

A Solid, **B** Liquid, **C** Gas

1 Melts (solid → liquid)

2 Evaporates (liquid → gas)

3 Freezes (liquid → solid)

4 Condenses (gas → liquid)

⭐ 8.5 How can Water be Evaporated Faster or Slower?

? **What are the conditions that affect how fast water evaporates? What differences do you see in evaporation on a cold day versus a hot day? (fan, drying cloth, sweating, windy day, hot day, rainy day)**

💡 SAMPLE ANSWER

Water evaporates faster on a **hot day** and slower on a cold day.

It also evaporates faster when there is **wind** or a fan, when the cloth is **spread out**, and slower on a **rainy (humid) day**.

Sweat dries fast under a fan or on a windy day.

Activity 8.7 **Take the same amount of water in a bottle cap and in a plate. Keep them near each other and record the time for complete evaporation (Table 8.4). What conditions did we keep the same? What did we change? What did we measure? What can you conclude?**

💡 **SAMPLE ANSWER**

Kept the same: amount of water, place (same sunlight, air, temperature).

Changed: exposed area of water.

Measured: time taken for the water to evaporate completely.

Exposed area of water	Time taken for complete evaporation
Less (bottle cap)	More time (e.g. about 3 hours)
More (plate)	Less time (e.g. about 1 hour)

Conclusion: The larger the exposed area, the **faster** the evaporation.

? **What would happen if milk is taken instead of water in the above activity?**

💡 **SAMPLE ANSWER**

The water in the milk would also evaporate faster from the plate than from the bottle cap.

But the milk would not disappear completely: a dry layer of milk solids would be left on the plate and the cap.

Table 8.5 **Design an activity similar to Activity 8.7 to find out other conditions that affect how fast water evaporates. What would you change? What would you keep the same?**

💡 **SAMPLE ANSWER**

Example: test the effect of moving air.

Kept the same: two identical plates with equal amount of water, same room.

Changed: one plate under a running fan, one away from the fan.

Condition changed	Time taken for complete evaporation
Under the fan	Less time
No fan	More time

Conclusion: Moving air makes water evaporate faster.

Activity 8.8 **Keep two identical bottle caps with equal water, one in sunlight and one in shade, and record the time to evaporate completely. What**

conclusions can you draw from this and other similar experiences?

✔ ANSWER

Water evaporates **faster in sunlight** (more heat) than in shade.

Clothes dry faster on a hot sunny day and on a **windy day** (more air movement).

Clothes dry slowly on a **rainy day**, as the air already has a lot of water vapour (more humidity).

? **If you want to dry your clothes on a rainy day, how can you make it faster?**

✔ ANSWER

Spread the clothes out well (more exposed area) and dry them under a fan or in a place with moving air.

Squeeze out extra water first, and use any heat available, such as ironing.

⭐ 8.6 Cooling Effect

? **Why is the water in the earthen pot (matka) so cold, but not in a stainless steel pot?**

✔ ANSWER

Water seeps out through tiny pores of the earthen pot and **evaporates** from its outer surface. Evaporation takes heat from the pot and the water, so the water becomes cool.

A steel pot has no pores, so no water evaporates from its surface and there is no cooling.

? What are the other examples of cooling effect?

💡 **SAMPLE ANSWER**

We feel cool under a fan as sweat evaporates.

Sprinkling water on the floor or roof in summer.

Wet cloth on the forehead during fever; a desert cooler; a surahi.

? How do you feel when you rub sanitiser on your hands?

✔ **ANSWER**

Our hands feel **cool**, because the sanitiser evaporates quickly and takes heat from our hands.

Activity 8.9 **Make a pot-in-pot cooler (two earthen pots, wet sand between them, wet jute sack or lid on top). Observe how it creates a cooling effect. For how many days can vegetables and fruits be kept**

fresh in it? What conditions can affect the number of days? What else can be used in place of sand for better cooling?

💡 SAMPLE ANSWER

Cooling: Water in the wet sand seeps through the outer pot and evaporates. Evaporation takes heat from inside, so the inner pot stays cool.

Vegetables and fruits stay fresh for several days (often about a week, depending on the vegetable).

Conditions: outside temperature, humidity, wind, keeping the sand wet, keeping the pot in shade.

Instead of sand: sawdust, coconut fibre (coir), charcoal pieces or wet cloth/jute, which hold water well.

⭐ 8.7 How do Clouds give us Rain?

Poem **Aavi enjoys the rain and creates a poem. Complete the poem (two missing lines after "How does rain fall?") and present it in your class.**

💡 SAMPLE ANSWER

I wonder, oh! I wonder so,

Which path does water choose to go?

I wonder, oh! I wonder so,

When does it snow?

I wonder, oh! I got a wonder call,

How does rain fall?

From seas and ponds it rises high,

And cools to clouds up in the sky.

I wonder, ponder and dream each day,

As water's journey takes its way.

Activity 8.10 **Squeeze and release a closed plastic bottle with a cup of water in it for 2–3 minutes. Repeat after adding a small burnt piece of newspaper. What will you observe?**

✔ **ANSWER**

Without the burnt paper, hardly anything is seen above the water.

With the burnt paper, some **haziness (a cloud)** appears above the water.

The burnt paper gives tiny dust particles. Water vapour condenses around them and forms a cloud. So dust particles help clouds to form.

Activity 8.11 **Label Fig. 8.9 using the words: Cloud, Lake, Ocean, River, Groundwater, Evaporation, Condensation, Rain, Snow — to show where water is stored, how it changes state and where it moves.**

💡 SAMPLE ANSWER

Where water is stored: Ocean, Lake, River, Groundwater (below the ground), Cloud (in the sky), Snow (on mountain tops).

Arrows: Evaporation — upward arrow from the ocean, lake and river to the sky.

Condensation — in the sky, where water vapour forms clouds.

Rain — arrows from the clouds down to land and water; Snow — falling on the mountain tops.

Rain water flows into rivers, seeps down as groundwater, and rivers carry it back to the ocean.

(check the picture in your book)

? **What did I do well? Was I able to label all the parts of the water cycle? Which parts of the water cycle were unclear to me?**

💡 SAMPLE ANSWER

Self-reflection, e.g.: I could label evaporation, condensation and rain easily. The movement of groundwater back to rivers and the ocean was less clear to me, so I read that part again.

Let us enhance our learning

1 **Which of the following best describes condensation?**

(i) The conversion of water into its vapour state.

(ii) The process of water changing from a liquid into gaseous state.

(iii) The formation of clouds from tiny water droplets.

(iv) The conversion of water vapour into its liquid state.

✔ ANSWER

(iv) The conversion of water vapour into its liquid state.

(i) and (ii) describe evaporation.

2 **Identify in which of the given processes evaporation is very important—**

(i) Colouring with (a) crayons (b) water colours (c) acrylic colours (d) pencil colours

(ii) Writing on paper with (a) pencil (b) ink pen (c) ball point pen

✔ ANSWER

(i) **(b) water colours**: the water in the paint must evaporate for the colour to dry.

(ii) **(b) ink pen**: the liquid ink dries on paper by evaporation.

3 **Space around natural grass feels cooler than space around plastic grass. Can you find out why?**

✔ ANSWER

Natural grass gives out water vapour from its leaves, and water in the moist soil below it evaporates too. Evaporation causes a **cooling effect**.

Plastic grass has no water to evaporate; it also gets hot in the sun. So it feels warmer.

4 **Give examples of liquids other than water that evaporate.**

✔ ANSWER

Hand sanitiser, perfume, spirit (alcohol), petrol, kerosene, nail-polish remover, deodorant.

5 **It might seem strange to use a fan to dry wet clothes since fans usually make things cooler, not warmer. Normally, when water evaporates, it requires heat, not cold air. What do you think about this?**

✔ ANSWER

A fan does not heat the clothes, but it **moves the air**. Moving air carries away the water vapour near the clothes and brings in drier air, so more water can evaporate.

The heat needed for evaporation comes from the clothes and the surroundings (that is why wet clothes feel cold). So clothes dry faster under a fan.

6 **Sludge removed from drains is left in heaps next to the drain for 3–4 days and then taken to a garden or field to be used as manure. This reduces transportation cost and makes it safer to handle. Reflect upon it and explain how.**

✔ ANSWER

In 3–4 days, much of the water in the sludge **evaporates**.

The sludge becomes lighter, so it costs less to carry; it also does not drip or spill.

Dry sludge is less sticky, smells less and is easier and safer for workers to handle.

7 **Observe the activities in your house for a day. Identify the activities that involve evaporation. How does understanding evaporation help us in daily activities?**

💡 SAMPLE ANSWER

Activities: drying clothes, drying the mopped floor, drying washed utensils, drying hair, boiling water or milk, making tea, cooling water in a matka, drying pickles and papad in the sun.

It helps us dry things faster (spread them out, use sun and wind), keep water cool (matka, surahi), keep ourselves cool (fan, cotton clothes in summer) and make salt from sea water.

8 **How is water present in the solid state in nature?**

✔ ANSWER

As **ice and snow** on high mountains and glaciers, ice at the North and South poles, frozen lakes and rivers in very cold places, and as **hail** that falls from clouds.

9 **Reflect on the statement "Water is our responsibility before it is our right." Share**

your thoughts.

💡 SAMPLE ANSWER

Everyone has the right to clean water, but only a small part of Earth's water is fit for use and the demand is growing.

So first we must do our duty: not waste water, close running taps, fix leaks, reuse water (e.g. for plants), collect rain water, and keep rivers and ponds clean. Only then can everyone, today and in future, get water.

10 **The seat of a two-wheeler parked on a sunny day has become very hot. How can you cool it down?**

✔ ANSWER

Sprinkle some water on it or wipe it with a wet cloth. The water **evaporates** and takes heat from the seat, so it cools down (cooling effect).

Parking it in the shade or covering it also helps.

📘 Learning further

1 **Wet one hand with water and leave the other dry. Blow air across both hands and feel the cooling effect. Find out the reasons for it.**

✔ ANSWER

The **wet hand feels cooler**.

Blowing air makes the water on the wet hand evaporate faster. Evaporation takes heat from the hand, so it cools. The dry hand has no water to evaporate.

2 **Make a game to navigate through different states of water and water related concepts to reach the finish line, with challenge cards on water cycle, evaporation, condensation, etc.**

💡 **SAMPLE ANSWER**

Sample: a snakes-and-ladders board with squares marked Ice, Water, Vapour, Cloud, Rain.

Challenge cards, e.g. "Name the process: vapour → water" (condensation). A right answer moves you forward (a ladder), a wrong one moves you back. The first to reach "Ocean" (finish) wins.

3 **Discuss with your teacher and act out the stages of the water cycle through a role-playing activity in your school assembly.**

💡 **SAMPLE ANSWER**

Sample: students play Sun, Ocean, Water drops, Cloud, Wind and Mountain.

> The Sun heats the Ocean, drops rise (evaporation), join hands to form a Cloud (condensation), the Wind takes them over the Mountain, they fall as rain or snow, and flow back to the Ocean by a river.

🎉 **Chapter 8 done! Revise the answers once more tomorrow.**